

HDOC DAY – 23

Aim

To write a simple C program that displays a menu to print different square and rectangular patterns using nested loops and switch-case statement.

Algorithm

1: Start.

2: Display the menu choices for pattern options:

1. Solid Square Pattern
2. Solid Rectangular Pattern
3. Row-Number Pattern
4. Column-Number Pattern
5. Exit

3: Read the user's choice.

4: Execute the corresponding function using a switch statement:

- **If Choice = 1:** Ask user for n (number of rows/columns). Use two nested loops to print * in n rows and n columns.
- **If Choice = 2:** Ask user for m (rows) and n (columns). Use nested loops to print * for m rows and n columns.
- **If Choice = 3:** Ask user for m (rows) and n (columns). Print row index i inside nested loops.
- **If Choice = 4:** Ask user for m (rows) and n (columns). Print column index j inside nested loops.
- **If Choice = 5:** Exit the program.
- **Default:** Print "Invalid Choice!".

5: Stop.

TEST CASE TABLE -

TEST CASE	INPUTS	EXPECTED OP
Case 1: Solid Square Pattern	Choice: 1, Rows/Cols (n): 3	*** *** ***
Case 2: Solid Rectangular Pattern	Choice: 2, Rows (m): 2, Cols (n): 4	**** ****
Case 3: Row-Number Pattern	Choice: 3, Rows (m): 3, Cols (n): 3	111 222 333
Case 4: Column-Number Pattern	Choice: 4, Rows (m): 3, Cols (n): 3	123 123 123

C CODE :

```
#include <stdio.h>
```

```
// Function for solid square pattern
```

```
void solidSquare() {
```

```
    int n, i, j;
```

```
    printf("Enter number of rows/columns (n): ");
```

```
    scanf("%d", &n);
```

```
    for (i = 1; i <= n; i++) {
```

```
        for (j = 1; j <= n; j++) {
```

```
            printf("*");
```

```
        }
```

```
        printf("\n");
```

```
    }
```

```
}
```

```
// Function for solid rectangular pattern
```

```
void solidRectangle() {
```

```
    int m, n, i, j;
```

```
    printf("Enter number of rows (m): ");
```

```
    scanf("%d", &m);
```

```
    printf("Enter number of columns (n): ");
```

```
scanf("%d", &n);  
for (i = 1; i <= m; i++) {  
    for (j = 1; j <= n; j++) {  
        printf("*");  
    }  
    printf("\n");  
}  
}
```

// Function for row-number pattern

```
void rowNumberPattern() {  
    int m, n, i, j;  
    printf("Enter number of rows (m): ");  
    scanf("%d", &m);  
    printf("Enter number of columns (n): ");  
    scanf("%d", &n);  
    for (i = 1; i <= m; i++) {  
        for (j = 1; j <= n; j++) {  
            printf("%d", i);  
        }  
        printf("\n");  
    }  
}
```

```
// Function for column-number pattern
```

```
void columnNumberPattern() {  
    int m, n, i, j;  
    printf("Enter number of rows (m): ");  
    scanf("%d", &m);  
    printf("Enter number of columns (n): ");  
    scanf("%d", &n);  
    for (i = 1; i <= m; i++) {  
        for (j = 1; j <= n; j++) {  
            printf("%d", j);  
        }  
        printf("\n");  
    }  
}
```

```
int main() {  
    int choice;  
  
    printf("--- MENU ---\n");  
    printf("1. Solid square pattern\n");  
    printf("2. Solid rectangular pattern\n");  
    printf("3. Row-number pattern\n");
```

```
printf("4. Column-number pattern\n");  
printf("Enter your choice: ");  
scanf("%d", &choice);  
  
switch (choice) {  
    case 1:  
        solidSquare();  
        break;  
    case 2:  
        solidRectangle();  
        break;  
    case 3:  
        rowNumberPattern();  
        break;  
    case 4:  
        columnNumberPattern();  
        break;  
    default:  
        printf("Invalid choice!\n");  
}  
  
return 0;  
}
```

CODE SCREENSHOT -

```
GW PICO 3.09

#include <stdio.h>

// Function for solid square pattern
void solidSquare() {
    int n, i, j;
    printf("Enter number of rows/columns (n): ");
    scanf("%d", &n);
    for (i = 1; i <= n; i++) {
        for (j = 1; j <= n; j++) {
            printf("*");
        }
        printf("\n");
    }
}

// Function for solid rectangular pattern
void solidRectangle() {
    int m, n, i, j;
    printf("Enter number of rows (m): ");
    scanf("%d", &m);
    printf("Enter number of columns (n): ");
    scanf("%d", &n);
    for (i = 1; i <= m; i++) {
        for (j = 1; j <= n; j++) {
            printf("*");
        }
        printf("\n");
    }
}

// Function for row-number pattern
void rowNumberPattern() {
    int m, n, i, j;
    printf("Enter number of rows (m): ");
    scanf("%d", &m);
    printf("Enter number of columns (n): ");
    scanf("%d", &n);
    for (i = 1; i <= m; i++) {
        for (j = 1; j <= n; j++) {
            printf("%d", i);
        }
        printf("\n");
    }
}

// Function for column-number pattern
void columnNumberPattern() {
    int m, n, i, j;
    printf("Enter number of rows (m): ");
    scanf("%d", &m);
    printf("Enter number of columns (n): ");
    scanf("%d", &n);
    for (i = 1; i <= m; i++) {
        for (j = 1; j <= n; j++) {
            printf("%d", j);
        }
    }
}
```

```

        }
        printf("\n");
    }
}

int main() {
    int choice;

    printf("--- MENU ---\n");
    printf("1. Solid square pattern\n");
    printf("2. Solid rectangular pattern\n");
    printf("3. Row-number pattern\n");
    printf("4. Column-number pattern\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch (choice) {
        case 1:
            solidSquare();
            break;
        case 2:
            solidRectangle();
            break;
        case 3:
            rowNumberPattern();
            break;
        case 4:
            columnNumberPattern();
    }
}

```

```
        case 4:
            columnNumberPattern();
            break;
        default:
            printf("Invalid choice!\n");
    }

    return 0;
}
```

OUTPUT SCREENSHOT :

```
2017: Command not found: 4
[manalithakar@Unknown_4e:f1:c3:0c:fe:82 ~ % ./23q1
--- MENU ---
1. Solid square pattern
2. Solid rectangular pattern
3. Row-number pattern
4. Column-number pattern
Enter your choice: 1
Enter number of rows/columns (n): 3
***
***
***
[manalithakar@Unknown_4e:f1:c3:0c:fe:82 ~ % ./23q1
--- MENU ---
1. Solid square pattern
2. Solid rectangular pattern
3. Row-number pattern
4. Column-number pattern
Enter your choice: 2
Enter number of rows (m): 2
Enter number of columns (n): 4
****
****
[manalithakar@Unknown_4e:f1:c3:0c:fe:82 ~ % ./23q1
--- MENU ---
1. Solid square pattern
2. Solid rectangular pattern
3. Row-number pattern
4. Column-number pattern
Enter your choice: 3
Enter number of rows (m): 3
Enter number of columns (n): 3
111
222
333
[manalithakar@Unknown_4e:f1:c3:0c:fe:82 ~ % ./23q1
--- MENU ---
1. Solid square pattern
2. Solid rectangular pattern
3. Row-number pattern
4. Column-number pattern
Enter your choice: 4
Enter number of rows (m): 3
Enter number of columns (n): 3
123
123
123
[manalithakar@Unknown_4e:f1:c3:0c:fe:82 ~ % clear
```